Hyunah Ko

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Education

- **Yonsei University** — Seoul, South Korea
  B.S. in Civil and Environmental Engineering — *Mar. 2022 – Present*
  Double Major: Computer Science — *GPA: 3.98/4.5*

Research Interests

My research goal is to develop advanced frameworks for 3D scene understanding and generation that enable (1) physically plausible 3D dynamic synthesis, (2) high-quality neural rendering with accurate material modeling, and (3) seamless integration with generative AI models. Particularly, my interests include:

- Physics-based 3D dynamic scene synthesis
- Diffusion models for 3D content generation
- Neural rendering and 3D Gaussian Splatting

Team Projects

- **Multi-View Diffusion Gaussian Splatting(MVDGS)***
  Developed novel view synthesis system combining CameraCtrl and 3D Gaussian Splatting to generate controllable 3D objects from single images with temporal consistency. *May. 2025 – Jun. 2024*
- **Recipe Recommendation via Object Detection and Knowledge Graph QA***
  Built a recipe recommendation system using YOLOv5 and knowledge-augmented prompts with Stable Diffusion image generation. *Oct. 2024 – Nov. 2024*
- **Pet Body Measurement via Depth Estimation**
  Developed a depth-based system for pet body length measurement using MMPose and fine-tuned Depth Anything v2 with patch-wise refinement. *Oct. 2024 – Nov. 2024*
- **Generative Pet Memorials with LLaMA3 and Stable Diffusion**
  Generated personalized pet letters and images using LoRA-finetuned LLaMA3 and Stable Diffusion with ControlNet and style transfer. *Aug. 2024 – Sep. 2024*
- **LLM-based Personalized Health Report Generation**
  Developed a personalized report generation system using medical LLMs and Retrieval-Augmented Generation. *Jul. 2024 – Sep. 2024*
- **Seoul Night Bus Route Analysis and Design***
  Analyzed public transportation and mobility data to identify underserved regions in Seoul. Applied EDA and regression modeling to propose new night bus routes based on demand patterns. *Jul. 2024 – Aug. 2024*
- **Room Generation with Stable Diffusion and ControlNet***
  Trained a customized model using ControlNet and Stable Diffusion to generate personalized room images from segmentation and depth data. *Apr. 2024 – May. 2024*
- **A(uto)ssignment: Automated Problem Extraction from Textbooks**
  Built an automated pipeline using YOLOv7 and PORORO OCR to detect and extract questions from textbook PDFs, enabling custom problem set generation. *Mar. 2024 – Apr. 2024*
- **Traffic Commute Analysis for Northern & Southern Seoul**
  Analyzed ridership and station data to identify high-congestion areas in commutes from Gyeonggi Province to Seoul using EDA and statistical methods. *Jan. 04, 2024 – Jan. 30, 2024*

* Served as a team leader.

Papers

- **Fourier Decomposition for Explicit Representation of 3D Point Cloud Attributes**
  Donghyun Kim, Hyunah Ko, Chanyoung Kim, Seong Jae Hwang, Under review.
- **Backbone Augmented Training for Adaptations**
  Jae Wan Park, Junhyeok Kim, Youngjun Jun, Hyunah Ko, Seong Jae Hwang, arXiv:2506.04288.

Research Experiences

- **Environmental Biotechnology & Thermal Laboratory (eBTL), Yonsei Univ.** Seoul, South Korea
  Undergraduate Researcher (Advisor: Prof. Joonhong Park) *Jul. 2023 - Sep. 2023*
- **Medical Imaging and Computer vision, Yonsei Univ.** Seoul, South Korea
  Undergraduate Researcher (Advisor: Prof. Seong Jae Hwang) *Jul. 2024 - Jul. 2025*
- **CVLAB, KAIST AI.** Seoul, South Korea
  Undergraduate Researcher (Advisor: Prof. Seungryong Kim) *Jul. 2025 - Present*

Relevant Courseworks

- **Spring 2023**
  - **UNDERSTANDING AND USING AI** : A+ (4.3/4.3)
- **Fall 2023**
  - **LINEAR ALGEBRA AND ITS APPLICATION** : A- (3.7/4.3)
- **Spring 2024**
  - **ARTIFICIAL INTELLIGENCE AND WATER INFORMATICS** : A- (3.7/4.3) *(G1)*
  - **DATA STRUCTURES** : A+ (4.3/4.3)
- **Fall 2024**
  - **INTRODUCTION TO AI AND ITS APPLICATION** : A+ (4.3/4.3)
  - **NATURAL LANGUAGE PROCESSING AND CHATGPT** : A+ (4.3/4.3)
  - **OBJECT-ORIENTED PROGRAMMING** : A0 (4.0/4.3)

Skills

- **Programming Languages**: Python (Expert), Java (Advanced), C++ (Intermediate), JavaScript (Beginner)
- **Development Tools**: Git, Docker, Linux/Unix, Matlab
- **Languages**: Korean (Native), English (Proficient)

Honors and Awards

- **Grand Prize, 2024 Jeju Satellite Data Utilization Competition**
  Proposed a data-driven solution and business model to address regional issues in Jeju Island by analyzing satellite imagery and identifying actionable insights. *Jul. 2024 – Aug. 2024*
- **77th out of 514, AirQo African Air Quality Prediction Challenge, Zindi**
  Developed a model to estimate air pollution levels using satellite imagery, addressing the high cost of ground-based sensor networks in large regions like Africa. *Mar. 2024 – Jun. 2024*

Extracurricular Activities

- **Yonsei Computer Club** *Sep. 2023 – Present*
  Vice President (Feb. 2024 – Feb. 2025)
- **Yonsei Artificial Intelligence** *Dec. 2023 – Present*
  Vice President (Jan. 2025 – Present)
- **Yonsei Data Science Lab** *Dec. 2023 – Nov. 2024*